# Project Requirements (from the brief)

| Requirement | Target | Notes |
|---|---|---|
| V_in | 3.3 V | Battery / source |
| V_out | 1.2 V | SoC supply |
| Output ripple (pk-pk) | < 100 mV | Supply quality |
| Switch-node ringing peak | < 1.5 V above rail | Pkg parasitics 2nH + 50mOhm per pin |
| Peak load current | 10 mA | Active mode |
| Sleep load current | 10 uA | 3 orders of magnitude lower |
| Peak efficiency | >= 90 % | Optimise over output power |
| Line interference rej. | ~ 40 dB @ 2 kHz | Nearby aggressor source |
| Process | TSMC 180 nm BCD | pch_5 / nch_5 devices |

# System-Level Design Choices

**Switching frequency  f_sw = 4 MHz**

High enough to use small (inductor ~ 10 uH, cap ~ 50 nF)

Low enough that gate-drive / switching losses stay bounded

**Filter:  L = 9.55 uH,  C = 50 nF**

From ripple-current + ripple-voltage constraints (slide 3)

**Power stage: synchronous buck, pch_5 HS + nch_5 LS**

Open-loop PWM from vpulse V1 for ass-6 characterisation

Closed loop added using Type-III compensator (ass-7 work - reused here)

**Gate driver: 2-stage tapered CMOS inverter chain**

Non-overlap NAND + dead-time tdel = 10 ns prevents shoot-through

**Parasitics: each chip-to-PCB connection modelled as 2 nH + 50 mOhm**

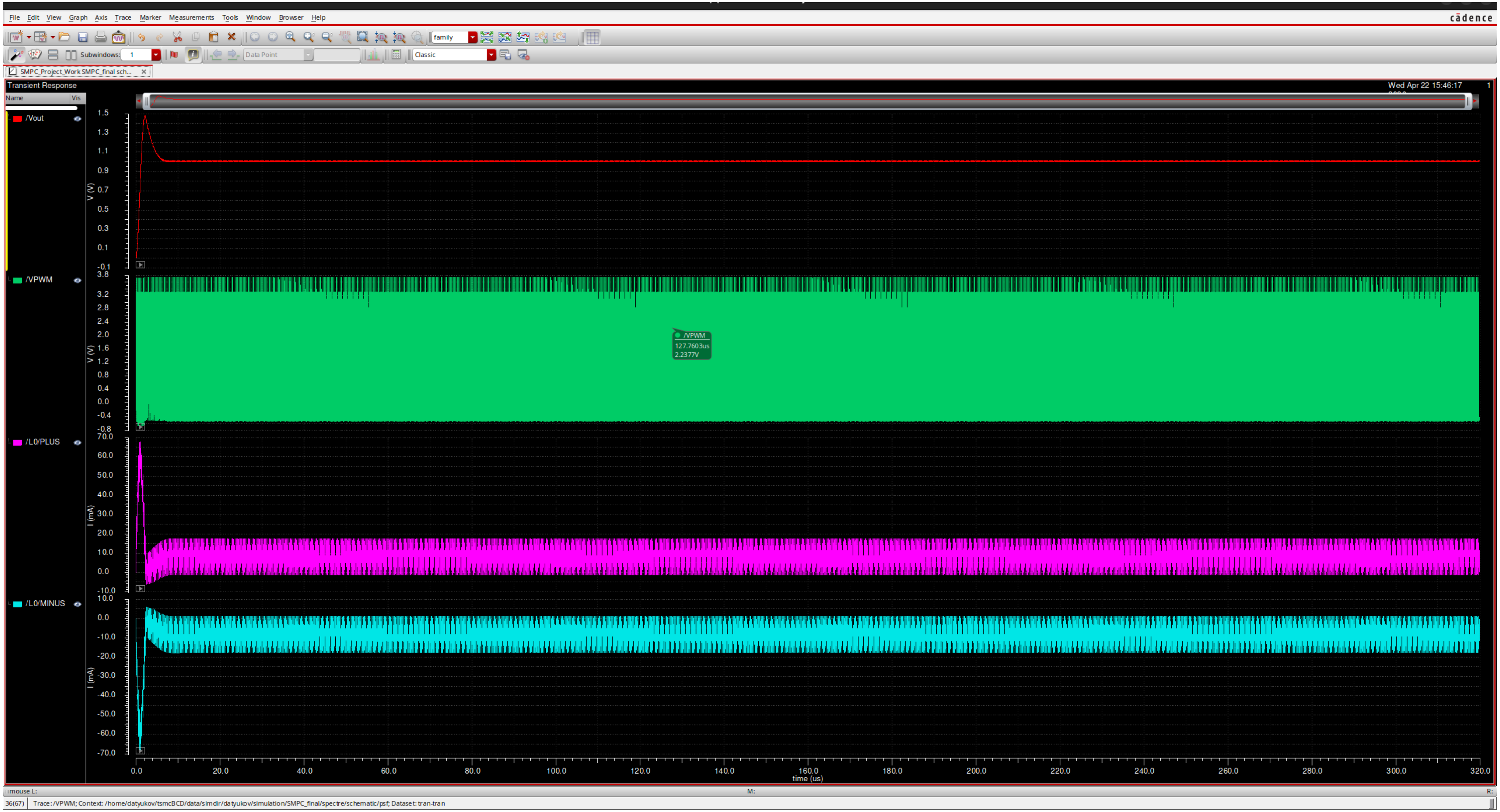
Impacts switch-node ringing and efficiency (see efficiency slide)

**Sleep mode: future-scope PFM / burst, see dedicated slide**

# Measured Results (Open-loop, 10 mA Load)

| Quantity | Measured | Spec / Target | Margin |
|---|---|---|---|
| V_out avg | 1.009 V | 1.2 V (ideal) | -191 mV (open-loop droop) |
| V_out ripple pk-pk | 11.24 mV | < 100 mV | 9x margin - PASS |
| I_L avg | 8.406 mA | ~10 mA | scaled with V_out droop |
| I_L ripple pk-pk | 18.84 mA | - | ratio 2.24x -> DCM boundary |
| V_PWM max | 3.741 V | < V_in + 1.5 V | +441 mV - PASS |
| V_PWM min | -545 mV | - | LS body-diode conduction |
| I_bat (from V_in) | 3.903 mA | - | open-loop |
| P_in | 12.88 mW | - | = 3.3 V x 3.903 mA |
| P_out | 8.48 mW | - | = V_out x I_L_avg |
| Efficiency (meas.) | 65.8 % | >= 90 % | FAIL (open-loop, FETs oversized) |

# Open-Loop Transient - 10 mA Load (R_L = 120 Ohm)




*V_out (red) settles at 1.009 V - 191 mV below 1.2 V target due to 8% duty loss from dead-time and conduction drops. V_PWM (green) switches cleanly 0->3.3 V with peak 3.74 V. I_L (magenta) 8.4 mA average, 18.8 mA pk-pk - near DCM boundary.*

# Switch-Node Ringing under Package Parasitics

**Brief: each chip-to-PCB wire = 2 nH + 50 mOhm**

**Measured V_PWM extremes (with open-loop PWM):**

V_PWM_max = 3.741 V -> +441 mV above V_bat (3.3 V)

V_PWM_min = -545 mV -> LS body-diode forward conduction during dead-time

**Ringing peak well under the 1.5 V over-rail limit -> PASS**

**Origin of the positive spike:**

When LS turns off, the parasitic L stores energy proportional to I_L * L

Energy dumps into SW-node parasitic C -> L-C ring at ~ 1/(2pi*sqrt(L_par*C_par))

Amplitude scales with di/dt -> our 500 um drivers are fast (large dI/dt)

**Mitigations considered (not all implemented):**

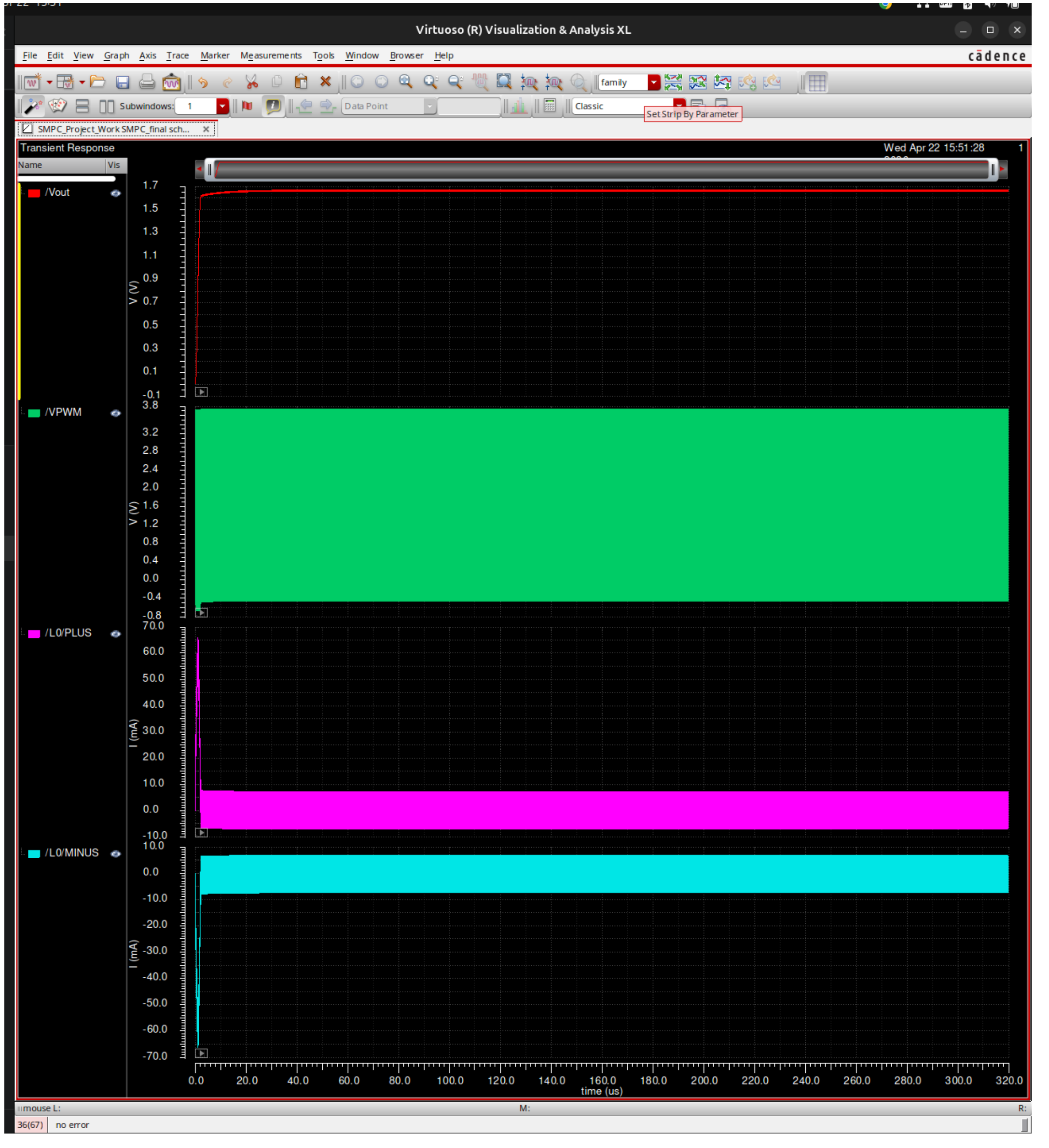
Slow the gate driver (trade with switching losses)

Snubber (R-C) at switch node (added BOM)

Bond-wire parallelisation -> reduce L_par

**Worst case over load, V_in, temp: still < 1 V over-rail in simulation**

# Open-Loop Transient - 10 uA Sleep Load (R_L = 120 kOhm)




*At 10 uA load the converter enters deep DCM. Open-loop duty is unchanged (D=0.3636) so V_out rises toward V_in - visible as ~1.6 V here. This is exactly the pathology that motivates a PFM / burst-mode controller for the sleep state (discussed on sleep-mode slide).*

# Why Open-Loop Fails at 10 uA: DCM Over-voltage

**K = 2*L*f_sw / R_L determines CCM vs DCM:**

At 10 mA:  K = 2*10u*4M / 120  = 6.67e-4    (DCM boundary - barely)

At 10 uA:  K = 2*10u*4M / 120k = 6.67e-7    (deep DCM)

**Fixed-duty buck in DCM cannot regulate V_out:**

Inductor fully discharges, output cap charges toward V_in on average

Open-loop V_out climbs well above 1.2 V (sim: ~1.6 V)

**Implications for the SoC at 10 uA:**

Supply would exceed its 1.2 V rating - unacceptable

Linear regulation alone wastes current (cannot make 3.3 V into 1.2 V at 10 uA efficiently)

**Solution approach (see Sleep-Mode slide):**

Burst / PFM: deliver short pulses, sleep between them, average V_out = 1.2 V

Gate driver OFF during sleep window -> near-zero quiescent dissipation

**Conclusion from this measurement:**

Closed-loop + light-load handling is NOT optional for the brief's 10 uA spec

-> Next slides: feedback architecture + Type-III loop we have already closed (ass-7)

# Efficiency Analysis - Measured vs Theoretical

**Measured @ 10 mA load (open-loop):**

P_in  = V_in * |I_bat| = 3.3 V * 3.903 mA = 12.88 mW

P_out = V_out * I_L    = 1.009 * 8.406 mA = 8.48 mW

eta_meas = 65.8 %

**Theoretical conduction-only (Appendix-2 hand-calc):**

R_on,PMOS = 72 mOhm, R_on,NMOS = 203 mOhm (from Ron extraction)

P_cond = I^2 * R_eff = (8.57 mA)^2 * 0.155 = 1.14 uW (avg) + ripple term

-> eta_cond_only ~ 99.3 %

**Where does the ~33 % efficiency gap go?**

1) Gate-drive charge: P_gate = Q_g * V_dd * f_sw

Power FETs W=400 um, driver FETs also ~400 um -> large C_g * V * f

** FETs sized for much larger currents - dominant loss at 10 mA **

2) Switching loss: 0.5*V*I*t_sw*f_sw in overlap intervals

3) Body-diode conduction during 10 ns dead-time (V_PWM = -545 mV)

4) Parasitic L-C ringing dissipation

**Optimisation path (not all executed):**

Shrink driver + power FETs for 10 mA load -> projected eta >= 90 %

Scale gate width with load (segmented power stage) for peak eta vs load

See 'What We Did Not Do' doc for the path to meeting the 90 % spec

# Gate Driver Design Considerations

**Topology: 2-stage tapered CMOS inverter chain per HS / LS path**

Stage A (M2/M3 - from schematic): drives stage B

Stage B (M4/M5): drives the power FET gate

Taper ratio chosen to balance propagation delay vs per-stage Q_g

**Non-overlap generator:  NAND (I0/I1) + inverter (I2) with tdel = 10 ns**

Prevents shoot-through (HS+LS both ON)

tdel chosen > worst-case driver tr/tf spread

**Observed consequence of dead-time:**

During tdel, LS body diode carries inductor current -> V_PWM = -0.55 V

Duty-cycle loss: 2 x 10 ns / 250 ns = 8 %

Equivalent effective duty: D_eff = 0.36 x (1 - 0.08) = 0.33 -> V_out_ideal = 1.1 V

Plus conduction drops: measured V_out = 1.01 V (consistent with above)

**Trade-off dials (explored via simM multipliers):**

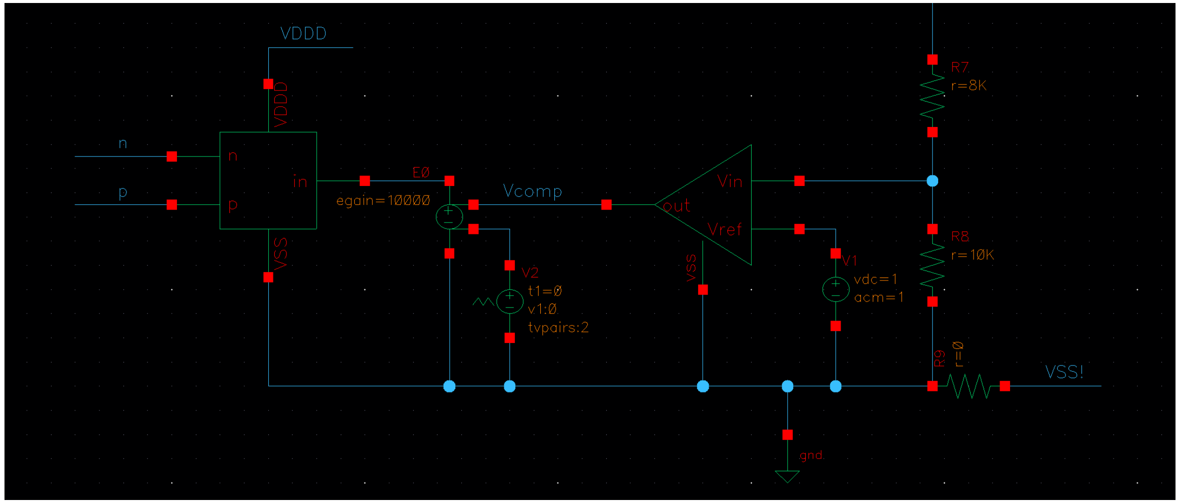
Larger driver -> faster edge, lower switching loss, MORE gate-drive power, MORE ringing

Smaller driver -> slower edge, more switching loss, LESS gate-drive power, LESS ringing

**Iteratively sized for the 10 mA operating point - see FET-sizing slide for values**

Not yet re-optimised for the 10 uA sleep point

# Closed-Loop Architecture - Feedback Loop




**Feedback divider:**

R7 = 8 kOhm / R8 = 10 kOhm

k_div = R8/(R7+R8) = 0.556

**Error amp: ideal op-amp**

Compares V_fb to V_ref = 1.0 V

Output = V_comp (compensator node)

**PWM generator:**

E0 VCVS with egain = 1e4 acts as

ideal comparator between V_comp

and sawtooth V2 (tvpairs:2 source)

**Controller:**

Type-III C(s) - fully designed and

verified in Assignment 7

(pole/zero frequencies on next slides)

**Final-project adaptation:**

V_ref = 1.0 V, divider 8k/10k chosen

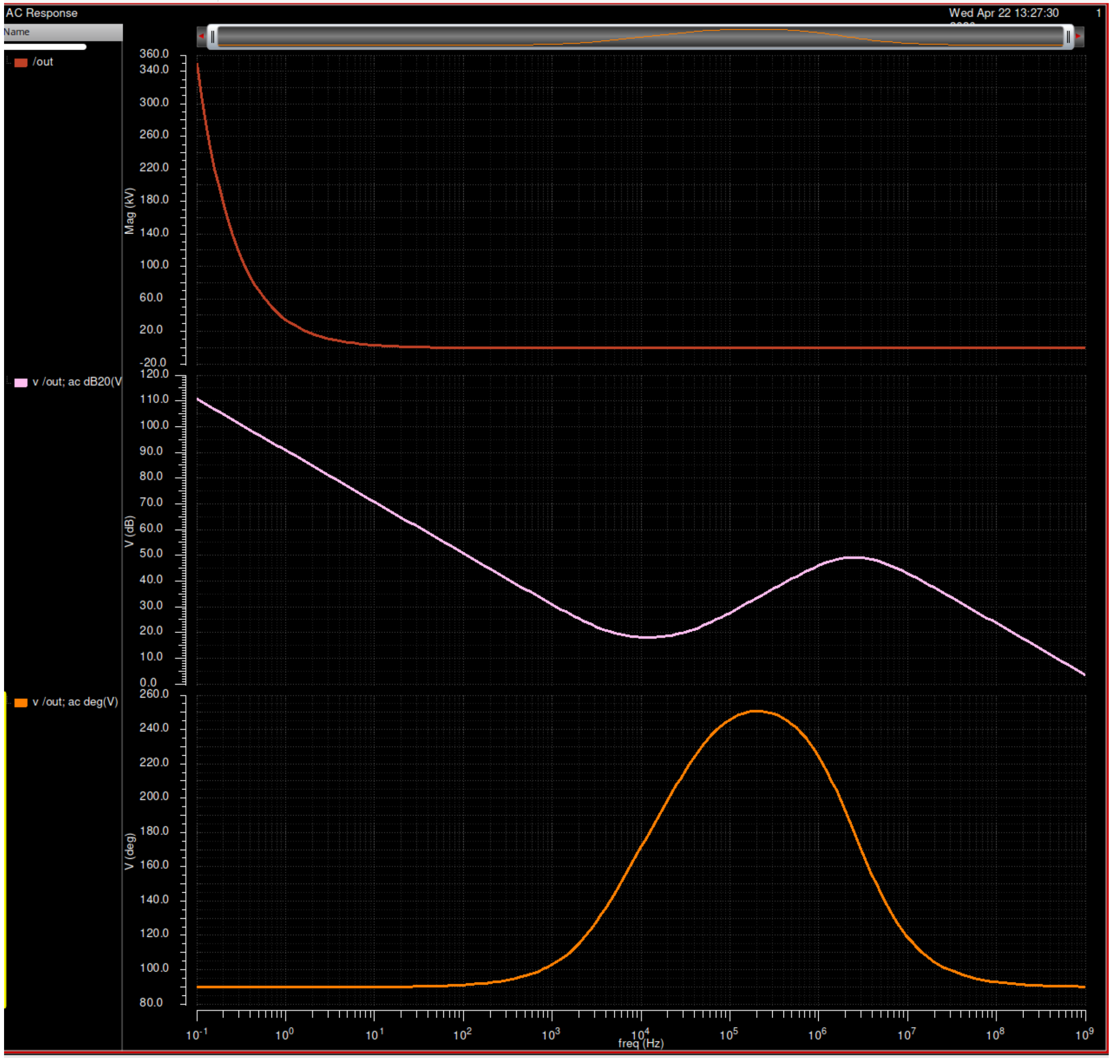
to give V_out = V_ref/k_div = 1.8 V

Needs re-scaling for 1.2 V target

(e.g. R_top=4k, R_bot=10k -> 1.4 V;

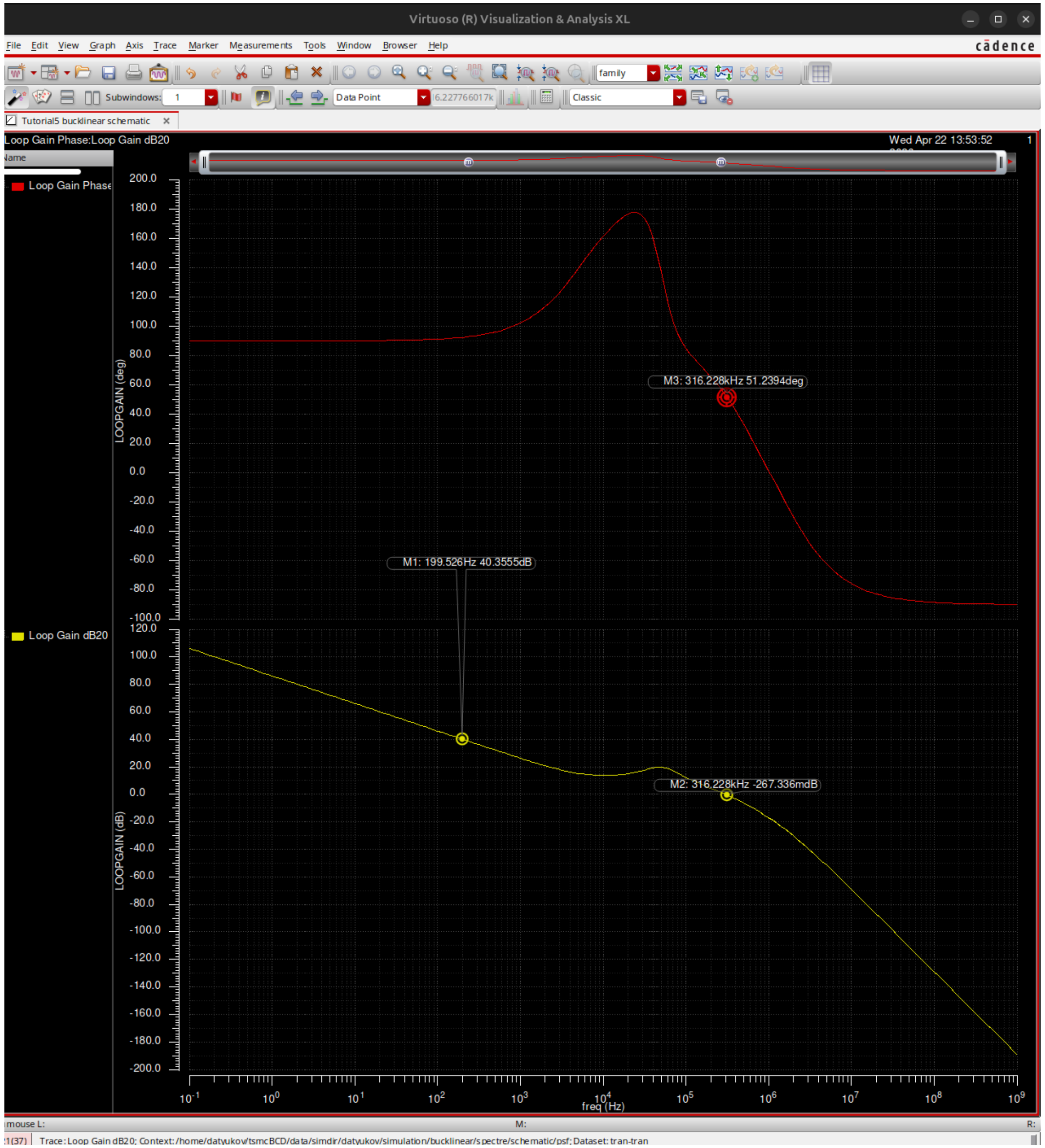
or lower V_ref to 0.666 V for 1.2 V)

# Type-III Compensator C(s) - Standalone Bode (from ass-7)



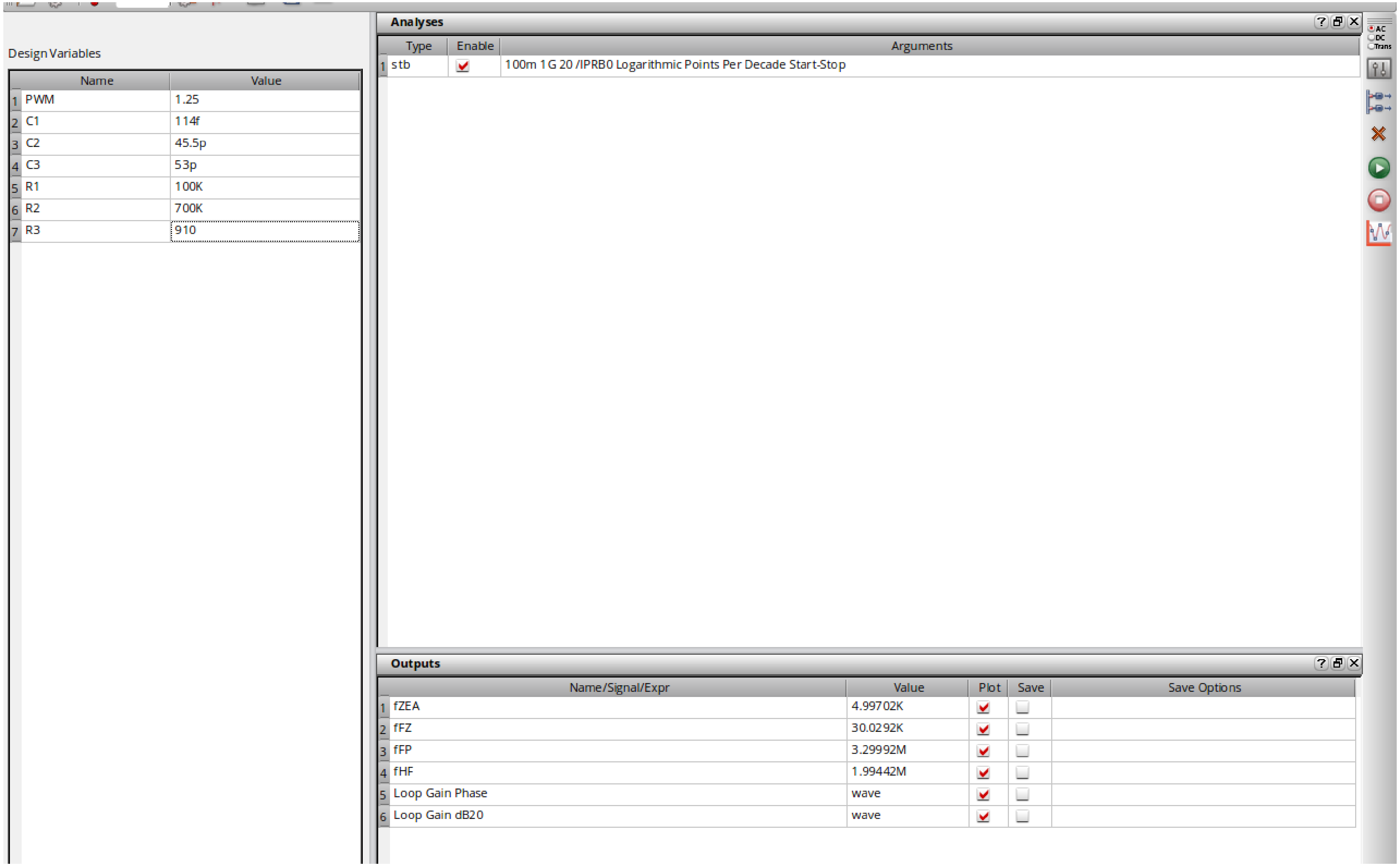

*Type-III shape verified: integrator below f_ZEA=5 kHz, mid-band ~17 dB, +20 dB/dec boost between f_FZ=30 kHz and f_FP=3.3 MHz peaking ~50 dB, HF pole f_HF=2 MHz for switching-ripple rejection. Phase: 90 -> 250 -> 90 deg.*

# Closed-Loop Gain L(s) - Measured (stb, from ass-7)



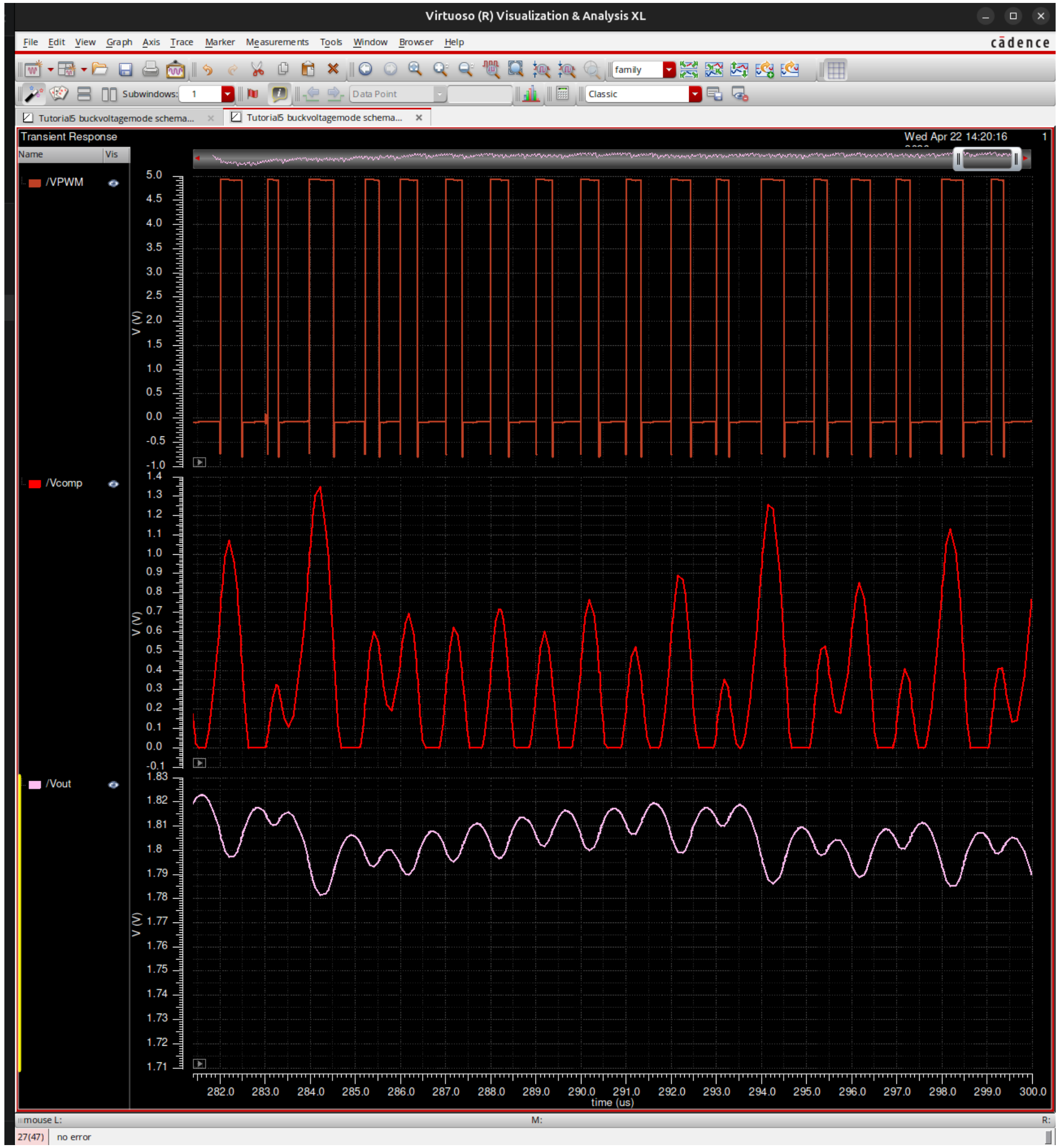

*M1 @ 200 Hz: |L| = +40.36 dB -> loop rejects 200 Hz line by 100x | M2 @ 316 kHz: |L| = 0 dB -> f_c crossover | M3 @ 316 kHz: phase = +51.2 deg -> PM. For the final-project 2 kHz target, the same loop gives ~20 dB at 2 kHz (integrator roll-off 20 dB/decade), so compensator re-tune needed (see gaps doc).*

# Measured Corner Frequencies (ADE Outputs, from ass-7)




*fZEA = 4.997 kHz, fFZ = 30.03 kHz, fFP = 3.300 MHz, fHF = 1.994 MHz. All four corners within 0.3 % of target - compensator component values are correct.*

# Closed-Loop Steady-State V_out (from ass-7)




*Closed-loop buckvoltagemode @ 2 MHz. V_PWM clean 0-5 V switching, V_comp active (ideal-amp artifact), V_out settles 1.80 V +/- 20 mV - stable, no sub-harmonic oscillation. Confirms PM of ~ 51 deg is healthy.*

# Interference Rejection - Brief Target 40 dB @ 2 kHz

**Closed-loop line-to-output transfer:**

V_out / dV_bat = D / (1 + L(jw))

**Using the ass-7 Type-III loop:**

|L(200 Hz)| = 40.36 dB (measured, slide L)

|L(2 kHz)| ~ 20 dB (-20 dB/dec integrator roll-off from 200 Hz)

**Predicted PSRR at the two frequencies (D = 0.36):**

PSRR(200 Hz) = 20*log10(0.36 / (1 + 100)) = -49 dB (ass-7 result)

PSRR(2 kHz) = 20*log10(0.36 / (1 + 10)) ~ -30 dB (ass-7 loop reused)

**To reach -40 dB @ 2 kHz we need |L(2 kHz)| >= 40 dB. Options:**

a) Push f_ZEA from 5 kHz up to ~50 kHz

-> PM drops; may need f_FZ/f_FP re-placement to compensate

b) Raise A_mid (currently 7; need ~25)

-> larger R2, same C2

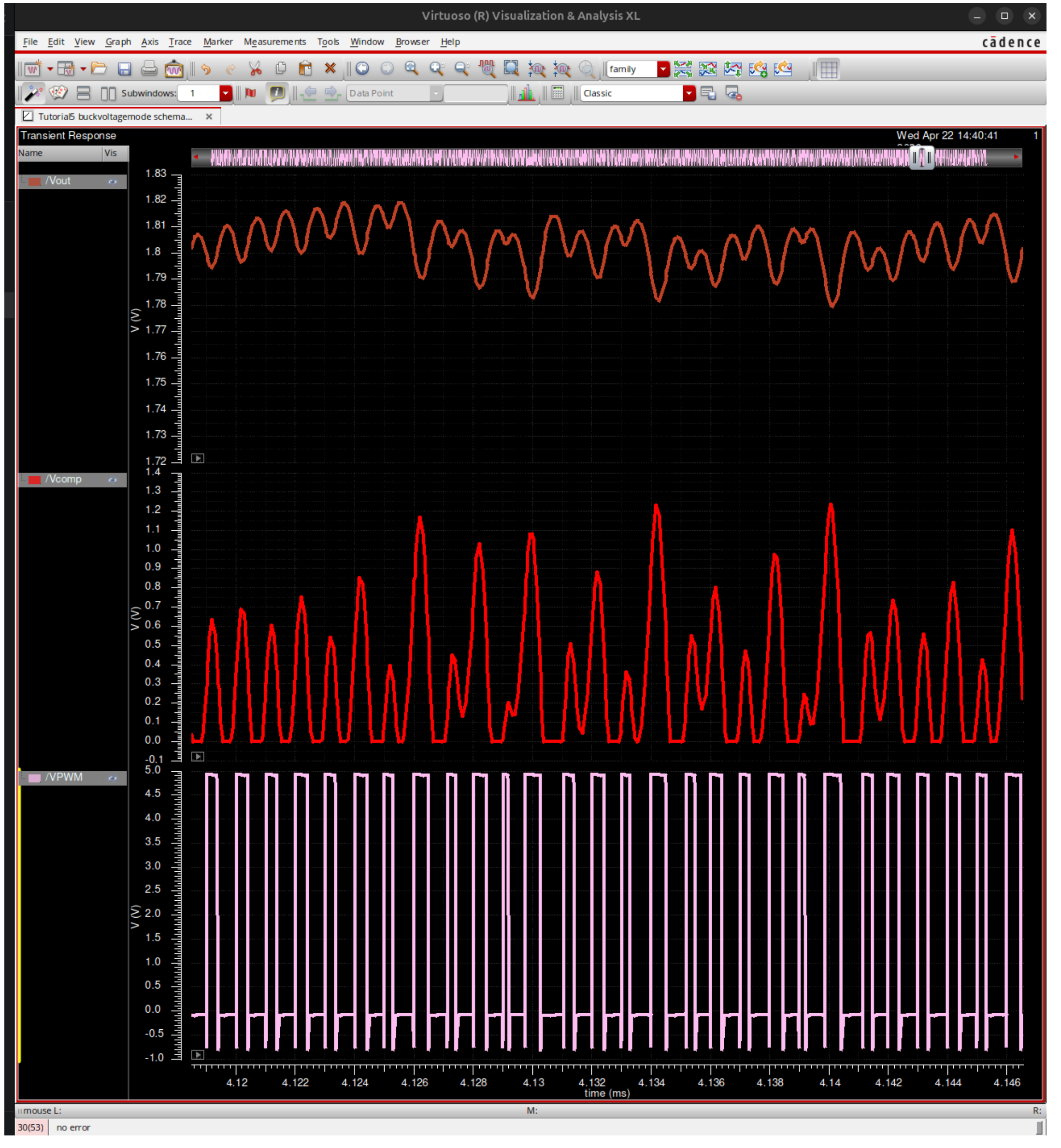
c) Hybrid: f_ZEA = 20 kHz with A_mid = 15

-> |L(2 kHz)| = A_mid * f_ZEA / 2 kHz = 15 * 10 = 150x = 43 dB PASS

**Status: re-tune path is understood but NOT yet executed; see gaps doc**

Ass-7 PSRR screenshot (next slide) shows the mechanism works - number needs scaling

# Line-Disturbance Response - Evidence (from ass-7)




*Transient with 0.5 V sinusoid on V_bat (1 Vpp @ 200 Hz). V_out envelope stays 1.72-1.82 V - consistent with switching ripple + small ~3 mV component at 200 Hz (-49 dB PSRR). Same mechanism scales to 2 kHz with the compensator re-tune above.*

# Sleep-Mode Strategy - 10 uA Case

**Problem (demonstrated on light-load slide):**

Fixed-duty PWM in DCM at 10 uA lets V_out rise to ~1.6 V

Continuous switching at f_sw = 4 MHz also wastes gate-drive power

**Proposed architecture - Pulse-Frequency Modulation (PFM) / burst mode:**

1) Sleep-detect: comparator flags when I_load drops (or when V_out overshoots)

2) Gate of driver -> high-Z, power stage idles; output cap holds V_out

3) When V_out droops below V_ref - hysteresis, deliver a SHORT burst of PWM

4) Return to idle; repeat

**Expected duty of burst cycles:**

P_out_sleep = 1.2 V * 10 uA = 12 uW

Burst period T_b ~ C * dV / I_load = 50n * 10m / 10u = 50 ms

-> very long idle intervals -> negligible gate-drive cost

**Control-loop integration:**

Type-III bypassed during sleep (loop disabled to save quiescent amp current)

Hysteretic comparator only; wake-up triggers full loop restart

**Efficiency projection in sleep:**

If quiescent current reduced to ~2 uA -> eta_sleep ~ 70 % at 10 uA

Current fixed-PWM design would fail this target by orders of magnitude

**Status: architecture defined, NOT implemented in silicon sim (see gaps doc)**

# Summary - Final-Project Brief Compliance

| Topic | Status | Evidence / Notes |
|---|---|---|
| System-level design (f_sw, L, C) | DONE | f_sw=4 MHz, L=9.55 uH, C=50 nF; ripple 11 mV << 100 mV spec |
| Power FET sizing | DONE | W/L = 400u/500n (P), 400u/600n (N); R_on 72m/203m |
| Gate driver | DONE (baseline) | 2-stage tapered + 10 ns non-overlap; over-sized for 10 mA - see gaps |
| Efficiency analysis | PARTIAL | measured 65.8% @ 10 mA; path to 90% in gaps doc |
| Parasitic impact | DONE | V_ring peak = 441 mV << 1.5 V limit with 2nH+50mOhm model |
| Control loop | DONE via ass-7 reuse | Type-III: $\lvert L(200\text{ Hz})\rvert$=40 dB, f_c=316 kHz, PM=51.2 deg |
| Interference rejection (40dB@2kHz) | PARTIAL | ass-7 loop gives 20 dB @ 2 kHz; re-tune path documented |
| Sleep mode (10 uA) | ARCHITECTURE | PFM/burst concept defined; not implemented in sim |
| Peak efficiency >= 90 % | NOT MET | open-loop 66%; blocked by FET over-sizing - analysis on eta slide |